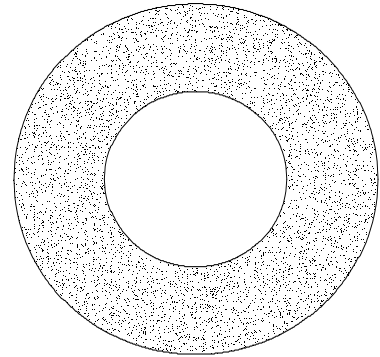


1. A coin is tossed until a head occurs. Write down the sample space of this experiment.
2. Two coins and a dice are tossed. Write down the elements of the event that the second coin falls heads.
3. A population has 4 elements: a , b , c and d . You will select 2 elements from this population at random with replacement. Write down the sample space of this experiment if the order of selection is important.
4. Two dice are thrown. Determine the probability that sum of the two numbers is seven.
5. Two dice are thrown. What is the probability that the range (larger value minus the smaller value) will be more than 3?
6. For two events A and B , we have $P(A) = 0.3$, $P(B) = 0.5$ and $P(A \cap B) = 0.1$. What is the probability that (a) either A or B occurs (b) A occurs but B does not and (c) neither A nor B occurs?
7. Refer to the Figure. The bigger circle has radius 2 and the smaller circle has radius 1. A particle is dropped such that it is equally likely to fall at any point inside the bigger circle. What is the probability that it will fall in the shaded area?



8. Probability that the extinguishing system works is 0.90, probability that the alarm system works is 0.70 and probability that both systems work is 0.65. Determine the probability that (i) at least one of the systems will be working? (ii) neither of the systems will be working?
9. Let $P(A) = 0.4$, $P(B) = 0.3$, $P(A \cap B) = 0.1$. Calculate $P(A | A \cup B)$.
10. For an unfair dice, $P(x) = \frac{x}{21}$; $x = 1, 2, 3, 4, 5, 6$. Calculate $P(3 | 3 \text{ or } 4)$.
11. Of flights from Saskatoon to Winnipeg, 89.5% leave on time and arrive on time, 3.5% leave on time and arrive late, 1.5% leave late and arrive on time, and 5.5% leave late and arrive late. What is the probability that, given that a flight leaves on time, it will arrive late? What is the probability that, given a flight leaves late, it will arrive on time?

12. Suppose that 40% of emails are spam. You want to design an “email filtering system” to deal with spam. You have prepared a list of “spam words” (words and phrases that are usually found in spam). If an email is spam, it contains spam words with probability 0.90. On the other hand, a regular email contains spam words with probability 0.05.
- What is the probability that a randomly chosen email contains spam words?
 - If the email contains spam words, what is the probability that it is spam?
13. If $A \subset B$, simplify (a) $P(A | B)$ (b) $P(A | B^c)$ and (c) $P(B | A)$.
14. A system functions if all of the four components A, B, C and D work. Each component works independently with probability 0.8. What is the probability that the system works?
15. The probability that a certain type of missile will hit the target on any one firing is 0.80. What is the minimum number of missiles that should be fired so that there is at least 0.99 probability of hitting the target at least once?
16. Information is transmitted digitally as a binary sequence known as ‘bits.’ However, noise on the channel corrupts the signal: a digit transmitted as 0 is received as 1 with probability 0.01, with a similar random corruption when the digit 1 is transmitted. It has been observed that, across a large number of transmitted signals, the 0s and 1s are transmitted in the ratio 2:3. A digit is transmitted. If the digit is received as 1, what is the probability that 1 was transmitted?
17. The probability mass function of X is presented below. Calculate (a) $P(X = 1 \text{ or } 13)$ and (b) $P(2 < X \leq 9)$.
- | | | | | |
|--------|-----|-----|-----|-----|
| x | 1 | 2 | 9 | 13 |
| $P(x)$ | 1/7 | 2/7 | 3/7 | 1/7 |
18. The density of X is given below. Determine the value of k , $P(X > 2)$, $P(0.7 < X < 2.7)$, density at $X = 2.0$ and $P(X = 0.5)$.
- $$f(x) = kx^3, \quad 0 < x < 4.$$
19. The density of X is given below. Determine the distribution function of X . Hence calculate $P(X \leq 1.5)$, $P(X < 0.4)$, $P(X > 0.6)$ and $P(0.5 < X < 1.7)$.
- $$f(x) = \frac{3}{8}x^2, \quad 0 < x < 2.$$
20. The cumulative distribution function of X is given below. Calculate $P(X \leq 5)$, $P(X > 6)$ and $P(5 < X \leq 7)$.

$$F(x) = \begin{cases} 0 & x \leq 2 \\ \frac{x-2}{7} & 2 < x < 9 \\ 1 & x \geq 9 \end{cases}$$

21. The joint probability function of X and Y is given below. Determine (a) the value of k (b) the marginal pmf of X and (c) conditional pmf of Y given $X = 0$.

$x \backslash y$	1	2	3
-1	$\frac{1}{17}$	$\frac{3}{17}$	$\frac{2}{17}$
0	$\frac{3}{17}$	k	$\frac{1}{17}$
1	$\frac{1}{17}$	0	$\frac{1}{17}$

22. The joint probability density function of X and Y is given below. Determine (a) the value of k (b) $P(0.2 < X < 0.7, 0.6 < Y < 0.9)$ and (c) the marginal density of X .

$$f(x, y) = \begin{cases} k(x + y) & 0 < x < 3, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

23. Refer to Question 21. Are X and Y independent?

24. Refer to Question 22. Are X and Y independent?

25. Probability function of X is given below. Determine the mean and the variance of X .

x	0	1	2	5
$p(x)$	0.1	0.2	0.4	0.3

26. The density of X is given below. Determine the mean and the variance of X .

$$f(x) = \begin{cases} x + 0.5 & 0 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$